

Question 1:

The system under consideration pertains to a network of automated banking machines (ABMs) designed to serve as decentralized kiosks, enabling bank customers to conduct cash deposits and withdrawals from their accounts. These ABMs are strategically distributed across various locations to provide convenient access to banking services for customers.

Operational functionalities of the ABM system include facilitating cash transactions such as deposits and withdrawals, wherein customers can interact with the machines to conduct financial transactions without the need for human intervention. This system caters to the core banking needs of customers

To do: Keeping in mind the above statement, **provide** an architecture diagram using the most suitable architecture styles. Clearly **mention/label** the architecture styles used. The diagram should be detailed enough to indicate the placement of system's modules. The diagram should not be a high level (and generic) diagram only. State your assumption(s) clearly.

Solution: Client-Server, Layering, Pipe and Filter or

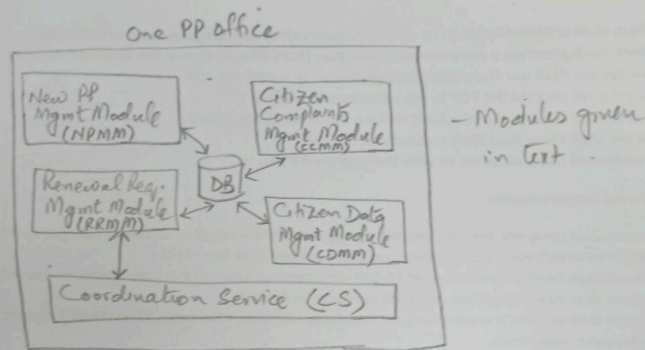
Peer-to-Peer, Layering, Pipe and Filter

Question 2:

Part a. The Passport department is planning to update its present architecture. Presently each authorized division keeps its own passport related data. There are 5 authorized divisions across the country. Each of these divisions is responsible for management of new passport requests, renewal of existing passports, citizen complaints, and citizen information updates. A citizen must visit an office for all services, for example they use kiosks in the divisional offices to record a complaint or apply for renewal etc. A citizen can request a new passport or information updates in the passports only at his/her birth division. The renewal requests can be placed at any office. In case of renewal requests, the passport office visited by the applicant contacts the concerned division (the division of birth of the applicant) through a coordination service running at the remote passport office (in the concerned division).

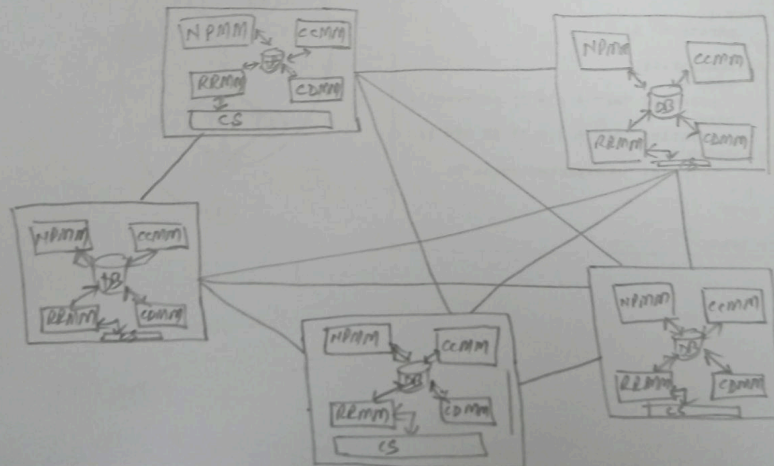
To do: Draw an architecture diagram expressing the present architecture of the passport office. Label the diagram completely naming the architecture styles used. The diagram should not be a general one, it should have complete detail of the parts of the system. **Justify** your choice of architecture style(s) using the text in the problem statement. The justification should not exceed 3 lines.

83 Part a



request placed at any office is sent to one concerned office, not the dedicated office

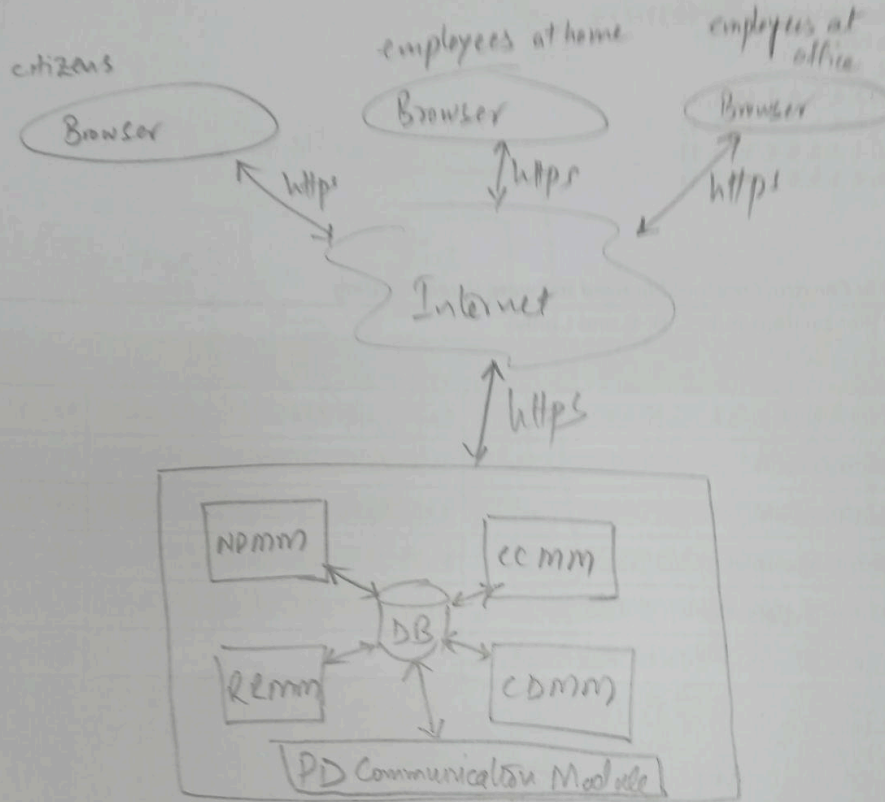
← P2P + Repo → keeps its own data



Part b. The plan now is to move towards a different architecture where citizens benefit and the department gets more control over applications and data. The oldest division (OD) shall keep all data and control over the passport applications. The OD will be responsible for communication between the other divisional offices. The citizens do not have to visit any divisional office now and can get all the existing services remotely.

To do: Draw an architecture diagram expressing the planned architecture of the passport department. Label the diagram completely naming the architecture styles used. The diagram should not be a general one, it should have complete detail of the parts of the system. **Justify** your choice of architecture style(s) using the text in the problem statement. The justification should not exceed 3 lines.

Q3 Part b

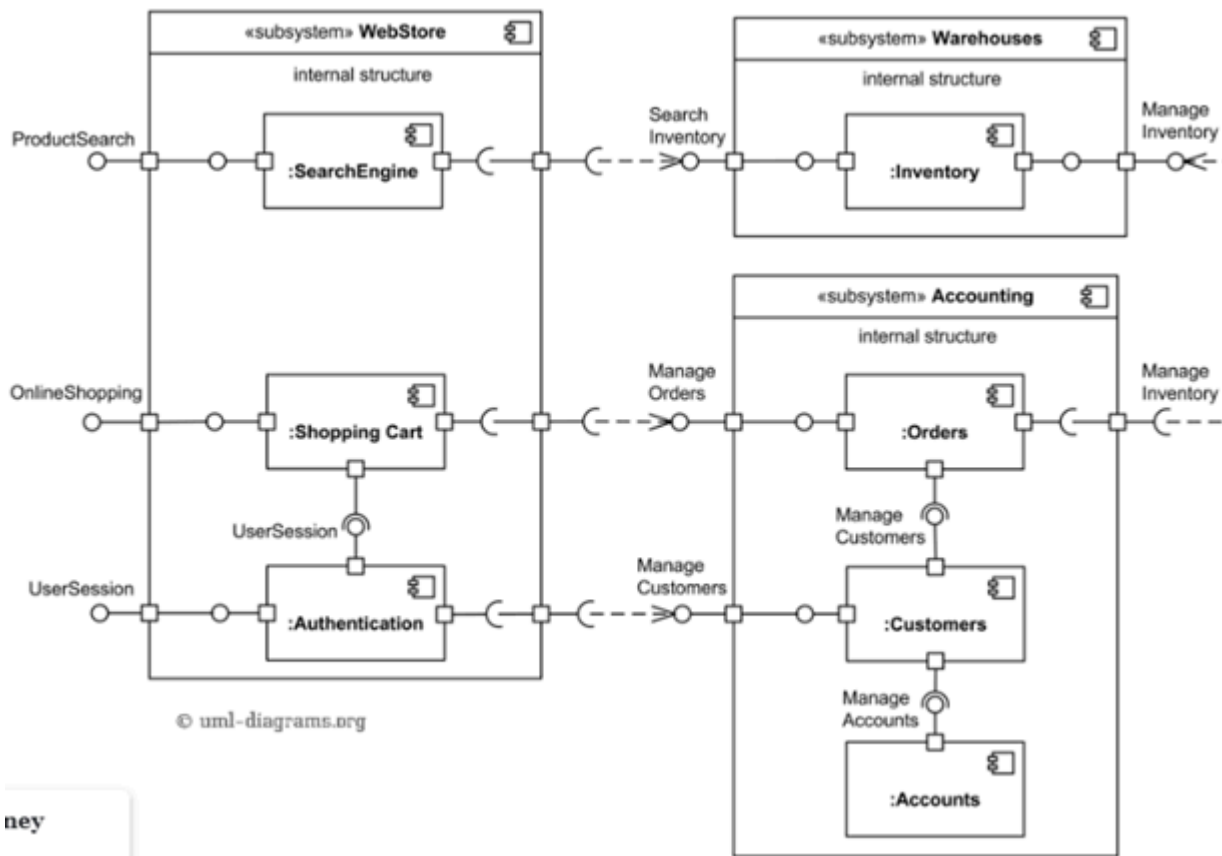


Min: C/S + Repo
 ↓ ↳ control over data
 One division
 maintains control
 over all data,
 app, communication

Question 3:

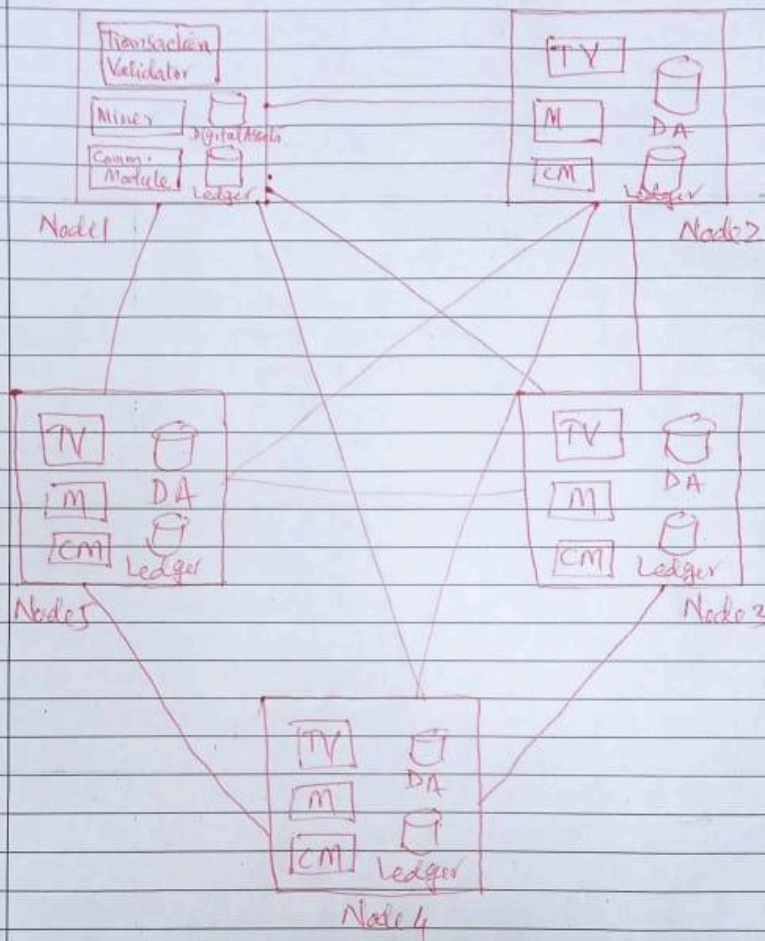
The WebStore system is designed to facilitate seamless online shopping by integrating three key elements: Search Engine, Shopping Cart, and Authentication. The Search Engine allows users to search or browse items through Product Search, which interacts with Search Inventory provided by the Inventory part of the Warehouses subsystem. This setup ensures accurate and up-to-date product information. The Shopping Cart leverages Manage Orders from the Orders part of the Accounting subsystem during the checkout process, ensuring that all transactions are processed efficiently and orders are correctly recorded. Meanwhile, the Authentication element enables customers to create accounts, login, and logout, binding each customer to their specific account to maintain security and personalized experiences. The Accounting module offers Manage Orders and Manage Customers, which are delegated to the Orders and Customers modules, respectively, ensuring robust order management and customer relationship handling. The Warehouses system's Search Inventory and Manage Inventory support these operations by maintaining the necessary product data and inventory levels, ensuring the WebStore subsystem operates smoothly and efficiently across all shopping activities.

Sketch a component diagram for the above system.



Q4: A cryptocurrency mining app (namely CryptoM) allows its users to grow their digital currency holdings directly through the mobile phones of the users. It also serves as a wallet to hold the **digital assets** of the users. This mobile app is energy-light and uses the resources as efficiently as possible. This app uses the **mobile devices of the users as edge devices** and allows the CryptoM company to avoid the need of high end computers and incur high electricity bills. This **user centric decentralized app involves nodes (smartphones) that communicate with each other** to **mine** the cryptocurrency securely. These nodes use a consensus algorithm to **validate transactions** and host a distributed **ledger** in parts. This distributed ledger (either whole or in parts) is available at multiple nodes. When requested by a particular node, the concerned node(s) share(s) the available version of the ledger with all concerned.

Suggest in one line the architecture style(s) that should be used for the architecture of CryptoM application. **Provide** an architecture diagram for the CryptoM application having 5 nodes actively involved in mining. **Label** the diagram completely and correctly. **Give reason(s)** for selecting the architecture style(s).

Architecture P2PReasons

- decentralized app
- all nodes communicate with each other
- No central or dedicated server mentioned or required.

Legend

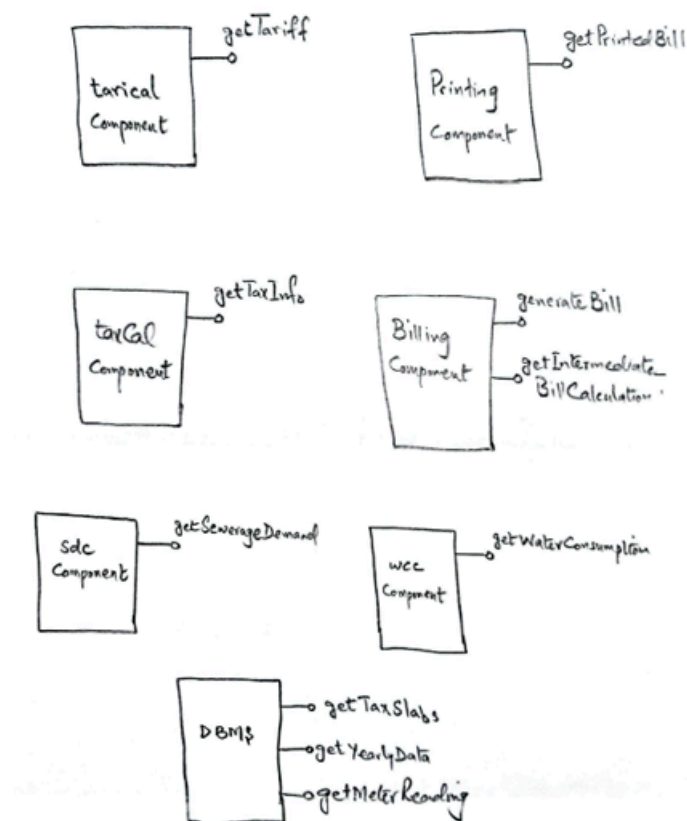
TV : Transaction Validator.

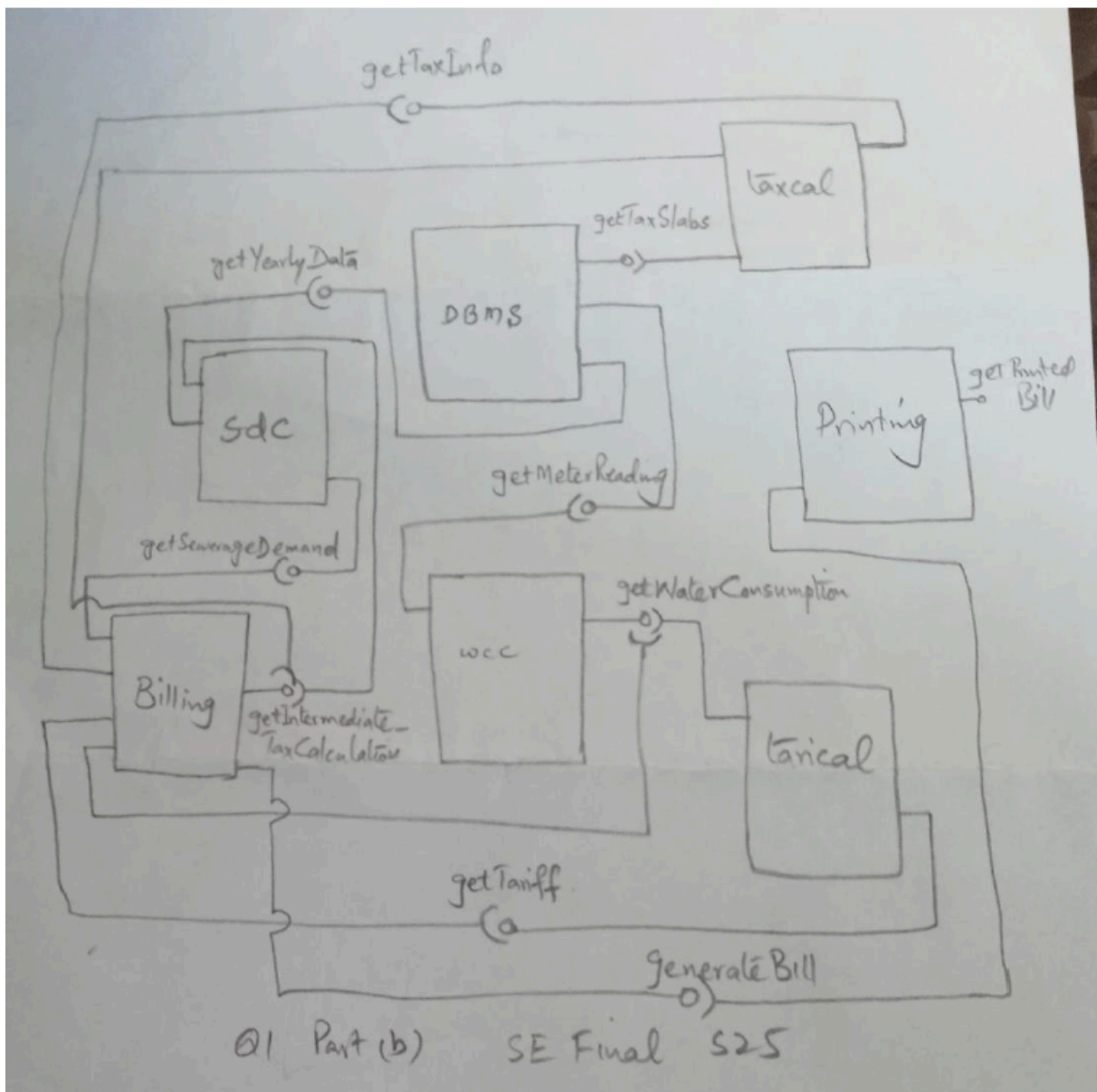
M : Miner

CM : Communication Module.

Q5: WASA has decided to computerize its whole system including the billing module. The billing module needs to interact with different other modules performing functions such as tariff calculation (tarical), tax calculation (taxcal), printing, sewerage demand calculation (sdc), and water consumption calculation (wcc). The main responsibility of the billing module is to generate bill, this bill is used by another module responsible for printing the bill. The billing module receives information from the different modules through their respective interfaces. The wcc module is responsible for providing the number of consumed units to other modules on request after receiving the water meter readings (or house size based estimations) from the DBMS. This information of consumed units is required by the tarical module that provides the tariff based on the slabs of water consumption. The tariff is operated with the units consumed by the billing module to get an intermediate value. This value is used by the taxcal and sdc modules to perform their respective responsibilities of calculating tax and calculating sewerage demand. The sdc module uses the DBMS provided yearly consumption data of the consumers to suggest any non-negative sewerage demand figure. The taxcal module uses the tax slabs available through the DBMS and the provided tax value is used by the billing module.

- Using the given information only, **design and provide** component interfaces (i.e. requires and provides interfaces) for all the components of the WASA system. **Use** ball and socket notation. Only show stand-alone components with their interfaces, do not connect the components with each other.
- Provide** a component diagram for the WASA system. **Use** ball and socket notation





Question 6)

One of the hospitals in the city plan to develop a Hospital Management System (HMS) to maintain the day –to-day state of admission/discharge of patients, list of available doctors, reports generation, attendance (including check-in and checkout) of staff, availability of beds in wards, availability of private rooms etc. The staff can be medical (s.a. doctors, nurses etc.) as well as non-medical (s.a. clerks, cashiers, receptionists, accountants etc.). Hospital allows administrator to view all the details of the hospital record. The admin can get details of any patient, doctor, ward, room, and staff. System shall store personal details and time/reason of visit, doctor visited for all patients who consult a doctor in hospital. The system shall keep complete history (previous visits, doctors consulted, medication, test report results, room/bed availed etc.) for all patients including outdoor patients as well as the indoor patients. The patients that do not get admitted but visit a doctor only are called outdoor patients and the patients that get admitted are called indoor patients. The patients can take appointment, through the HMS, from doctors online as per their availability. Each doctor will be assigned two specific days of the week for outdoor patients. System will also maintain lab records.

Following are some other objectives of the HNS:

1. To computerize all details regarding patient details & hospital details.
2. To schedule the appointment of patients with doctors to make it convenient for both.
3. To schedule the services of specialized doctors properly so that facilities provided by hospital are fully utilized in effective and efficient manner.
4. To keep the medical store's medicines inventory updated in case of issue and return of medicine
5. To handle records of the test reports prepared in the pathology lab of the hospital.

Provide functional decomposition of the HMS.

Provide functional decomposition of the HMS.

